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A liquid drop present on a solid surface can move because of a gradient in wettability along the surface. Such motion is known to occur in the direction of increasing wettability. In phase change heat transfer devices employed in reduced gravity, this phenomenon can be used to enhance the heat transfer rate by moving freshly formed condensate away and exposing the solid surface to the vapor, as noted by Daniel, Chaudhury, and Chen (Science, vol. 291, 633, 2001). Gradients of wettability also can be used to move small drops from one place to another in devices on a chip known as MEMS.
A systematic experimental study of the motion of drops on treated silicon/silicon oxide surfaces will be conducted and a theoretical analysis of this problem will be performed so that the predicted velocities of the drops can be tested against those measured experimentally. A wettability gradient can be produced on a silicon/silicon oxide surface by exposing it to an alkylsilane. The result is a strongly hydrophobic region at one end where the equilibrium contact angle can be as large as 100 degrees and a hydrophilic region at the other end where the contact angle is close to 0 degrees. By using a suitable preparation technique, contact angle hysteresis can be kept minimal, thereby facilitating motion of the drops. In the experiments, drops of various liquids in the size range 0.3 to 1.0 mm will be videotaped as they move along a silanized surface using a video camera equipped with a magnifying lens attachment, and their motion will be subsequently analyzed using image analysis software. Also, the shapes of the drops will be determined.

Task Description:

In analyzing problems involving moving contact lines, one must deal with a stress singularity at the contact line that arises when the classical hydrodynamic model is used. Different methods have been devised to eliminate this singularity, the most common one being the accommodation of slip in a region of molecular dimensions at the contact line. Use of this approach is justified by results from molecular dynamics simulations. The key parameters in the model are suitably defined Reynolds, Capillary, and Bond numbers, and a ratio of the slip length to the macroscopic dimension of the drop. In many of the proposed experiments, the Reynolds and Capillary numbers are expected to be small. Therefore, the initial theoretical efforts will focus on asymptotic analysis. The motion of a drop under conditions when inertia is negligible will be analyzed by solving the governing equations for the case where the ratio of the slip length to the dimensions of the drop is small. This analysis will be performed for the case when the Capillary number is small so that the shape of the drop will be only slightly different from that when it is static. To conduct a systematic experimental study of the motion of drops on surfaces with a gradient in wettability, and to perform a theoretical analysis of this problem so that the predicted velocities of the drops can be tested against those measured experimentally. The motion of drops in wettability gradients can be important in phase change heat transfer devices employed in reduced gravity, where it can be used to enhance the heat transfer rate by moving freshly formed condensate away and exposing the solid surface to the vapor. Gradients of wettability also can be used to move small drops from one place to another in devices on a chip known as MEMS, and to remove debris away from the vicinity of ink-jet nozzles in ink-jet printers.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Efforts during the reporting period were concentrated on obtaining data, developing theoretical predictions, and interpreting the data using the theory; also efforts were spent on obtaining and interpreting predictions from the molecular dynamics simulations. One article was published in Langmuir, and another was submitted for publication in the same journal. A third article was submitted for publication in the Proceedings of an IUTAM Conference. An oral presentation based on the molecular dynamics simulations was made in October 2004 at an IUTAM Symposium at Argonne National Laboratory, another on the same topic at the November 2004 Annual AIChE meeting in Austin, Texas, and two oral presentations based on results from the experiments and the molecular dynamics (MD) simulations were made at the 79th ACS Colloid and Surface Science Symposium held in Potsdam, New York in June 2005. Results from experiments and theory
The experiments were performed by first preparing a wettability gradient on a thin strip of silicon approximately 30 mm x 20 mm. The strip was cut out of a polished silicon wafer, and subjected to a cleaning procedure described in the last annual report. The strip was wide enough to permit experiments to be performed on separate tracks for each drop in a set of approximately 10 drops of varying size. In most cases, drops of the same size were permitted to migrate on separate non-adjacent tracks to check for reproducibility of the data. The liquid used for the experiments was tetraethylene glycol, which yielded reproducible data, and therefore was chosen over ethylene glycol, which had been used in prior experiments. The nominal volume of the drops was varied from approximately 50 to 2500 nanoliters. Data were obtained from seven different strips, but only four of these yielded usable data over a wide range of drop sizes. Three different gradients, labeled gentle, intermediate, and sharp, were employed.

When a suitable gradient was formed on the strip, it was placed on a stage mounted on a precision positioning device. The device permitted the stage to be moved in the three coordinate directions by small distances in a precise manner so that the drop could be brought into the field of view of an imaging system that had been assembled. The motion of the drop was observed and recorded using a set of digital video cameras equipped with magnifying lenses. One of these cameras recorded its plan view as it moved, while another captured a side view. The digital images were directly acquired at the rate of 50 frames per second by a computer for later analysis. The trajectory of each drop was analyzed from the recorded video images using Spotlight (software provided by NASA). From this information, the velocity of each drop as well as its footprint radius were obtained as functions of time. The instantaneous velocity at a given position was calculated as the local slope of a straight line fitted to a set of points chosen symmetrically about the selected position of the center of the drop’s footprint on the position versus time curve. The number of points was chosen judiciously to minimize the error associated with scatter in the data if too few points are taken, and the error associated with possible curvature in the data if too many points are chosen.

In separate experiments performed on the tracks at the two ends of the strip, a small drop was placed at different locations along the gradient, and its images were recorded in high magnification. These images were analyzed to obtain the contact angle as a function of position along the strip as follows: Software that detects the edge (CANNY) was first used to find the outline of the drop shape. Then, the shape near the contact line was fitted to a third degree polynomial to extract the contact angle, using the method described by Bateni et al. (2003), who also provided guidelines on how best to obtain the contact angle from digitized images.

The drops typically reached a peak velocity in the first 2-3 mm of traverse, with the velocity subsequently decaying by approximately one order of magnitude in the next several mm of traverse. Larger drops move more rapidly in the same wettability gradient. To interpret the data, a quasi-steady theoretical model was developed. In this model, the instantaneous velocity of the drop is predicted by balancing the driving force acting on the drop against the

Task Progress:

hydrodynamic resistance offered to its motion by the solid surface. For predicting the resistance, two approaches were used. In the first, the drop is assumed to be divided into a series of wedges and the drag experienced by each wedge was obtained from a solution available from Cox (1986). In the second approach, an exact solution for the drag was obtained using the lubrication approximation. In both models, the drops were assumed to be spherical caps. This is consistent with the experimental observations made of drop shapes, because the deformation of shape due to gravity as well as that due to motion were both very small in the experiments. In the theoretical work, asymptotic results also were developed from the predictions for the case when the contact angle is small, as well as for the case when the ratio of the slip length (region near the contact line where the usual stress singularity is relieved using a slip model) to the drop's footprint radius is small. Results from the theory were published in Langmuir during 2005.

When the theoretical results from the experiments using the intermediate gradient were compared with the experimental data on drop velocity versus position, the qualitative trend of the data was correctly predicted by theory, and the best performer was the wedge model; however, quantitative differences remained. After considering a variety of explanations for these differences, it was found that an approach that accommodated contact angle hysteresis satisfactorily explained the discrepancies. This explanation is based on the fact that drops of sufficiently small size were found to remain stationary on the gradient surface, or move so slowly that they could be considered stationary for practical purposes. In fact, drops of such small sizes were employed for the contact angle measurements. Using the experimental data on the velocity of drops of different footprint radii plotted against the footprint radius, it was possible to extract the critical size at which a drop would just fail to move on the surface as a function of position along the gradient. Postulating that the net driving force (accommodating hysteresis) on this critical-size drop must be zero, a method was developed for reducing the driving force on drops of larger sizes than the critical size. This modified driving force was balanced against the hydrodynamic resistance to obtain the predicted velocity, which was found to agree well with the measured velocity, as a function of position along the gradient for drops of different sizes. Also, the same technique was used for interpreting the experimental data obtained from other surfaces with sharp and gentle wettability gradients, and found to work reasonably well. The results from the experiments and the interpretation using quasi-steady theory have been summarized in an article submitted for publication in Langmuir.

Results from molecular dynamics simulations

It has been postulated that molecular mechanisms that are incompletely understood at present are operative at the contact line in offering resistance to the motion of the contact line (Blake and Haynes, 1969, Brochard-Wyart and de Gennes, 1992). To elucidate the molecular phenomena occurring at the contact line, molecular dynamics simulations of drops moving in a wettability gradient have been performed. The simulations are based on ideas developed by Thompson et al. (1989, 1993), Yang et al. (1992), and de Ruijter et al. (1999). The principal results from these simulations are described here.

The simulations involve both the spreading of a drop on a homogeneous surface and the movement of a drop on a wettability gradient surface. The gradient is simulated by varying the energy of interaction between fluid and solid atoms linearly with distance. For the spreading experiments, the drop is formed above the surface, and moved toward the surface by the use of a body force. When the drop is near the surface, the body force is removed and the drop is permitted to spread. From snapshots of the positions of the atoms, or the density field, the shape of the drop is followed as a function of time. The contact angle at any given time is evaluated by fitting the shape of the drop to a spherical cap, excluding the first 10 layers of liquid atoms. The theory of Blake and Haynes (1969) is used to interpret the spreading data by fitting a friction coefficient and an equilibrium contact angle.

In the simulations of the spreading of drops on homogeneous surfaces, it was found that the friction coefficient formulation of Blake and Haynes provides an accurate parameterization of the motion of advancing contact lines. de Ruijter et al. (1999) reached the same conclusion. A difference between the present work and that of de Ruijter et al. is that the present results cover a broad range of surface wettabilities and have been obtained for receding contact lines as well as advancing contact lines. The friction coefficient for a receding contact line is found to be significantly larger than that for an advancing contact line. The Blake-Haynes theory makes no distinction between the friction coefficients for advancing and receding contact lines. In the initial calculations of the friction coefficients, viscous effects were ignored by analogy with de Ruijter et al. (1999); however, estimates of the importance of viscous effects suggest that they should be considered when computing friction coefficients. Although the values of the friction coefficients should be different when viscous resistance is included, the difference between the friction coefficients for advancing and receding contact lines should be unaffected. Once the viscosity of the liquid is computed from the simulations, the friction coefficients will be recalculated including the viscous resistance, which will be obtained using the wedge solution near the contact line from Cox (1986).

In runs involving a drop moving in a wettability gradient, the drop indeed migrates along the surface at a speed that is approximately of the order of magnitude predicted by a quasisteady model in which the driving force for motion is balanced against the resistance to motion offered at the contact line, ignoring the hydrodynamic drag. Results also have been obtained for drops migrating on surfaces composed of atoms that are either twice as large or half as large as the fluid atoms. In all cases, the velocity of the drop decreases dramatically as it moves onto more hydrophilic portions of the surface even though the driving force is approximately constant. When the wall atoms are half as large as the fluid atoms, the drop moves significantly faster than in the other simulations. This may be consistent with the experimental findings from this work that drops composed of relatively large molecules move significantly faster than drops composed of smaller molecules.

A paper was submitted for publication in the Proceedings of the IUTAM Symposium "Computational Approaches to Disperse Multiphase Flow". The title of the paper is "Molecular dynamics simulations of drop motion on uniform and non-uniform solid surfaces". The Symposium was held at Argonne National Laboratory in October 2004. The referee reports were received in August 2005. Both reports were positive. Minor revisions were made in response to comments made by the referees, and the paper was re-submitted.

Work is proceeding on a manuscript for submission to Langmuir. The viscous corrections will be included in the calculations of the friction coefficients. By the time the manuscript is completed, the MD simulation work will be brought to a close.

Plans for the Future

This is the final year of the project. A no-cost extension has been obtained to complete the experimental work and revise the article submitted for publication when the reviews become available, and to complete and submit a manuscript on

	the molecular dynamics simulations. These tasks will be completed during the coming months. References Allen, M. P. and Tildesley, D. J. 1987. Computer simulation of liquids, Oxford Science Publications, Oxford. Bateni, A., Susnar, S.S. , Amirfazli, A., and Neumann, A.W. 2003. A high-accuracy polynomial fitting approach to determine contact angles. Colloids and Surfaces A: Physicochem. Eng. Aspects 219, 215-231. Blake, T. D. and Haynes, J. M. 1969. Kinetics of Liquid/Liquid Displacement. J. Colloid Interface Sci. 30, No. 3, 421-423. Brochard-Wyart, F. and de Gennes, P. G. 1992. Dynamics of partial wetting. Adv. Colloid Interface Sci. 39, 1-11. Cox, R. G. 1986. The dynamics of the spreading of liquids on a solid surface. Part 1. Viscous flow. J. Fluid Mech. 168, 169-194. de Ruijter, M. J., Blake, T. D., and De Coninck, J. 1999. Dynamic wetting studied by molecular modeling simulations of droplet spreading. Langmuir 15, 7836-7847. Thompson, P. A. and Robbins, M. O. 1989. Simulations of contact-line motion: slip and the dynamic contact angle. Phys. Rev. Lett. 63, 766-769. Thompson, P.A., Brinckerhoff, W.B., and Robbins, M.O. 1993 Microscopic studies of static and dynamic contact angles. J. Adhesion Sci. Technol. 7, No. 6, 535-554. Yang, J., Koplik, J., and Banavar, J. R. 1992. Terraced spreading of simple liquids on solid surfaces. Phys. Rev. A. 46, 7738-7749.
Bibliography Type:	Description: (Last Updated: 12/04/2005)
Articles in Peer-reviewed Journals	Subramanian, R.S.; Moumen, M.; McLaughlin, J.B. "Motion of a Drop on a Solid Surface Due to a Wettability Gradient" Langmuir, 25, 11844-11849 , Dec-2005
Presentation	McLaughlin, J.B. ; Saravanan, S. S.; Moumen, N. ; Subramanian, R.S. "Molecular Dynamics Simulations of Drop Motion on Uniform and Non-Uniform Solid Surfaces" IUTAM Symposium on “Computational Approaches to Disperse Multiphase Flow”, Argonne National Laboratory, Illinois. Oct-2004
Presentation	McLaughlin, J.B.; Saravanan, S. S.; Moumen, N.; Subramanian, R.S. "Molecular Dynamics Simulations of Drop Motion on Surfaces with Wettability Gradients" AIChE 97th Annual Meeting, Austin, Texas. Nov-2004
Presentation	Moumen, N. ; Subramanian, R. S.; McLaughlin, J. B. "Motion of Liquid Drops on a Horizontal Surface with a Wettability Gradient" 79th ACS Colloid and Surface Science Symposium, Clarkson University, Potsdam, New York. Jun-2005
Presentation	Saravanan, S. S. ; McLaughlin, J. B.; Subramanian, R.S. "Molecular Dynamics Simulations of Nanodrop Motion on Uniform and non-Uniform Surfaces" 79th ACS Colloid and Surface Science Symposium, Clarkson University, Potsdam, New York Jun-2005